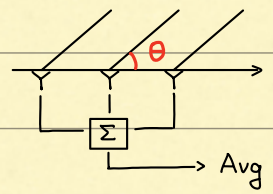
Beamforming　DOA 中，设 $K=1$

对 y 求和

$$G(\theta) = \frac{1}{M}\left|\sum_{m=1}^{M} y_m\right| = \frac{1}{M}\left|\sum_{m=1}^{M} e^{j\pi r_m \cos\theta}\right| = \left|\frac{\sin(\frac{1}{2}\pi M\cos\theta)}{M\sin(\frac{1}{2}\pi\cos\theta)}\right|$$

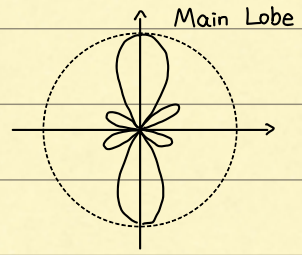


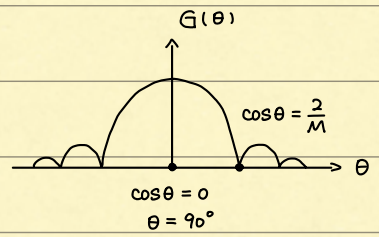
证：对于 ULA，等比数列

$$\sum_{m=1}^{M} e^{j\pi r_m\cos\theta} = 1\cdot\frac{1-e^{j\pi M\cos\theta}}{1-e^{j\pi\cos\theta}}$$

$$\sin\alpha = \frac{1}{2j}(e^{j\alpha}-e^{-j\alpha})$$

$$= \underbrace{\cancel{\frac{e^{j\cdot\frac{1}{2}\pi M\cos\theta}}{e^{j\cdot\frac{1}{2}\pi\cos\theta}}}}_{\text{模为 }1}\cdot\frac{e^{j\cdot\frac{1}{2}\pi M\cos\theta}-e^{-j\cdot\frac{1}{2}\pi M\cos\theta}}{e^{j\cdot\frac{1}{2}\pi\cos\theta}-e^{-j\cdot\frac{1}{2}\pi\cos\theta}} = \frac{\sin(\frac{1}{2}\pi M\cos\theta)}{\sin(\frac{1}{2}\pi\cos\theta)}$$





$\theta=0$ 即垂直入射，干涉最大

$x = \frac{1}{2}\pi\cos\theta$　当 $\theta=\frac{\pi}{2}$，$x=0$

$$G = \lim_{x\to 0}\frac{\sin Mx}{M\sin x} = \lim_{x\to 0}\frac{M\cos Mx}{M\cos x} = 1$$

$\sin(\frac{1}{2}\pi M\cos\theta)=0$，$\frac{1}{2}\pi M\cos\theta=\pi$ 为最近零点

带宽 $BW = 2\arccos\frac{2}{M}$

M 越多，主瓣越窄

对 y 加权求和

CB（Conventional Beamforming）/ Delay-Sum

$$\hat{\theta}_{CB} = \arg\max\ a(\theta)^H\hat{R}a(\theta)$$

权重 $w=a(\theta)$，称直角窗

等价于 $\min\|R-\hat{R}\|_F^2$，$K=1$

Pf：Noiseless $K=1$，$R = p\,a(\theta)a(\theta)^H$

$$\|R-\hat{R}\|_F^2 = p - a(\theta)^H\hat{R}a(\theta)$$

故 $a(\theta)^H\hat{R}a(\theta)$ 越大越可能真实 p

MVDR / Coupon

$$\hat{\theta}_{MVDR} = \arg\min\frac{1}{a(\theta)^H\hat{R}^{-1}a(\theta)}$$

权重为最大化 w^Hy 的功率

$$\max\ w^H\hat{R}w\quad \text{s.t. } w^Ha(\theta)=1,\qquad w = \frac{\hat{R}^{-1}a(\theta)}{a(\theta)^H\hat{R}^{-1}a(\theta)}$$

等价于 $\min\|R^{-1}-\hat{R}^{-1}\|_F^2$